

Vehicle Recognition for Car Listing Verification

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Motivation

Problem Motivation

- Used-car listings rely on seller-reported make/model information
- Incorrect or missing metadata hurts search quality
- Image-text mismatch creates listing-quality and fraud risk
- Manual make/model selection adds friction for sellers

Our Solution

- Build an image-based classifier to identify vehicle make/model
- Use the model as an independent verification signal
- Support cleaner listings, better search, and faster listing creation

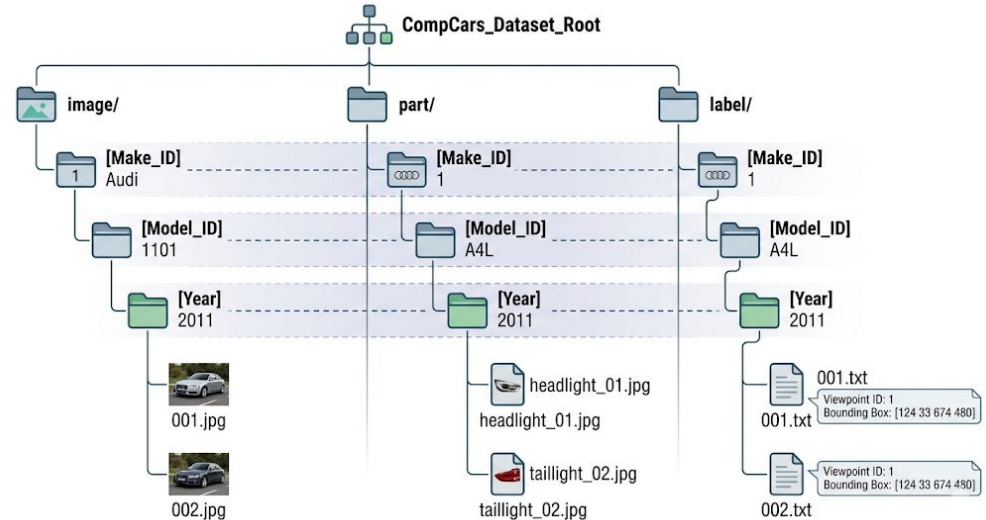
Stakeholders

- Marketplaces: cleaner metadata and fraud-screening signal
- Buyers: more trustworthy search results
- Sellers: faster listing creation through auto-fill

Data description

Dataset overview

- **Source:** CompCars Web-Nature subset, originally released by CUHK MMLab; accessed through a Kaggle mirror
- **Raw structure:** `image / make_id / model_id / year / xxx.jpg`
- **Input / predictor:** one vehicle image
- **Preprocessing:** resized to **299×299** with letterbox padding
- **Label / outcome:** composite `make_id/model_id` class



Data Decision

- Official split issue: covers only 431 / 1,706 make-model pairs
- Problem: official split would discard many usable vehicle classes
- Fix: scan the full filesystem before applying filters
- Class filters: keep make-model pairs with **≥ 3 production years** and **≥ 120 images**
- **Reasoning:** ≥ 120 images ensures enough samples per class; ≥ 3 years adds year-level visual variation
- **Filtered pool:** 406 classes, 52,490 images
- Model-specific splits: v4ab model uses 60/10/30 train-val-test; W model uses 70/15/15
- Used fields: vehicle images + make/model labels; no part-level images or labels used
- Note: v4ab uses whole images; W additionally uses provided bbox crops

Shared Architecture: Common Foundation

Same transfer-learning backbone and head skeleton used by both models



Important Design Choices and Hyperparameters



W: Top-1 95.15%, Top-5 99.21%,
Make Accuracy 95.99% | **v4ab baseline:** Top-1 91.99%,
Top-5 98.04%

Important Design Choices



Single-task vs Multi-task

- v4ab: single 406-class classifier
- W: separate make + model heads
- Better fit for marketplace verification



BBox Crop vs Full Image

- v4ab: no bbox crop
- W: ground-truth bbox-assisted cropping
- Raised performance ceiling
- But deployment is less direct without bbox labels



Data Filtering + Class Density

- Kept models with ≥ 3 years and ≥ 120 images
- Filtered set: 406 model classes
- Stronger per-class support
- Improved stability and generalization



Split Strategy

- Official split kept fewer usable classes
- Custom stratified split retained all filtered classes
- Better class balance
- More consistent training



Augmentation Design

- No augmentation $\rightarrow$ severe overfitting
- Too strong augmentation $\rightarrow$ underfitting
- Moderate augmentation worked best

Important Hyperparameters



Input Size (TARGET_SIZE)

- 224 $\rightarrow$ 299
- v4: Val Top-1 73.29%, Test Top-1 72.90%
- v4a: Val Top-1 82.62%, Test Top-1 84.02%
- Higher resolution preserved grille, headlights, and body-shape cues



Augmentation Strength

- No aug: Train 99.67%, Val 46.29% (gap 53.4%)
- v3 strong RandAugment + RandomErasing 0.5: Val 56.81%
- v4/v4a moderate factor 0.3 performed much better
- Best tradeoff between fit and generalization



MixUp (MIXUP_ALPHA)

- v4 no MixUp: Val 73.29%, Test 72.90%
- v5 $\alpha = 0.2$: Val 69.48%, Test 71.30%
- MixUp reduced accuracy, so we dropped it



Bottleneck + Fine-tuning Depth

- v3: bottleneck 256, unfreeze 154 $\rightarrow$ Val 56.81%, Test 59.54%
- v4: bottleneck 512, unfreeze 100 $\rightarrow$ Val 73.29%, Test 72.90%
- Larger bottleneck + deeper unfreezing improved adaptation
- Not a pure ablation because augmentation also changed



Final W Tuning

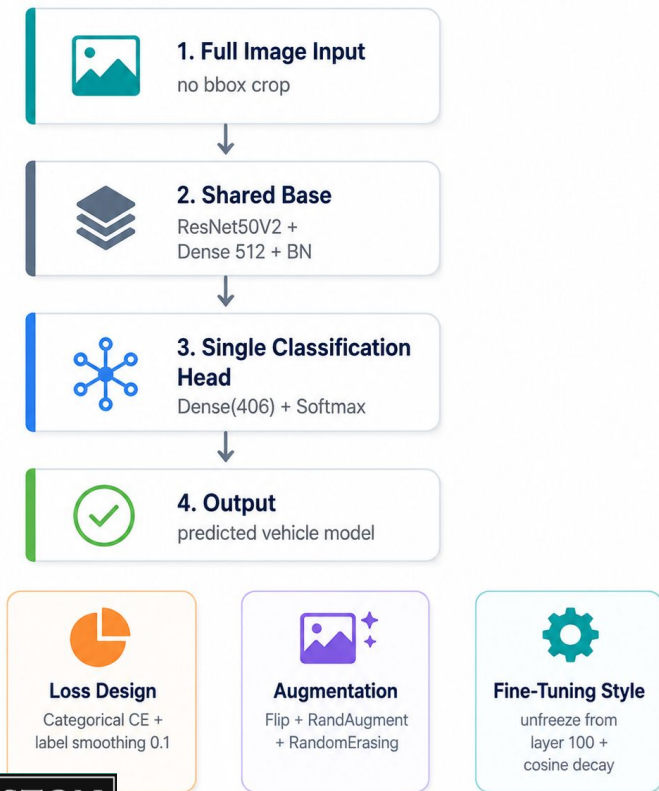
- hidden_dim 512, dropout 0.35
- warmup 4, finetune 36, unfreeze 140
- finetune_lr 5e-5
- make/model loss weights = 0.30 / 0.70
- Emphasized the harder, more important model task



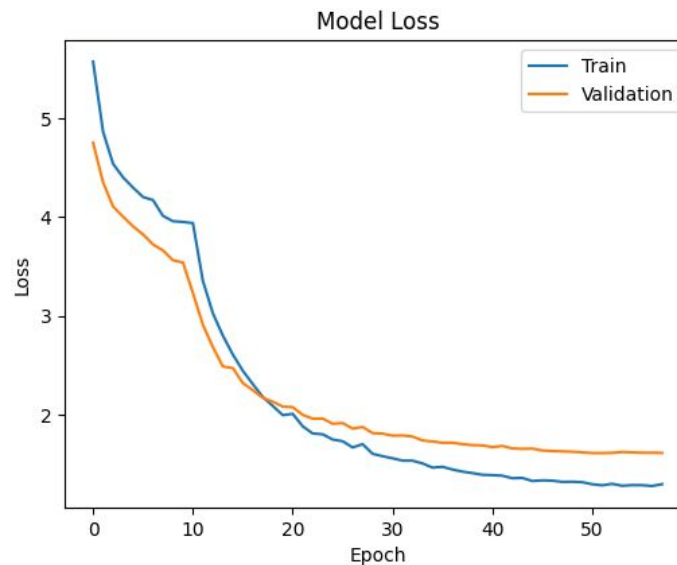
Takeaway: The biggest gains came from better input resolution, stronger data curation, and a shift from a single-task classifier to a bbox-assisted multi-task verification model.

v4ab-Specific Architecture

Single-task 406-class model classifier
with stronger regularization

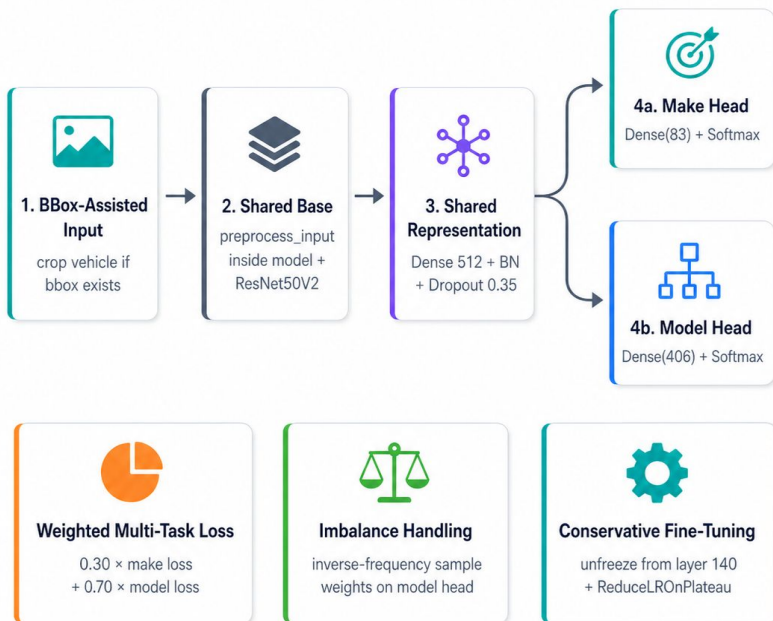


Output	Classes	Random Top-1	Test Top-1	Loss
v4ab model head	406	1/406 = 0.25%	Acc = 91.99%	1.4387

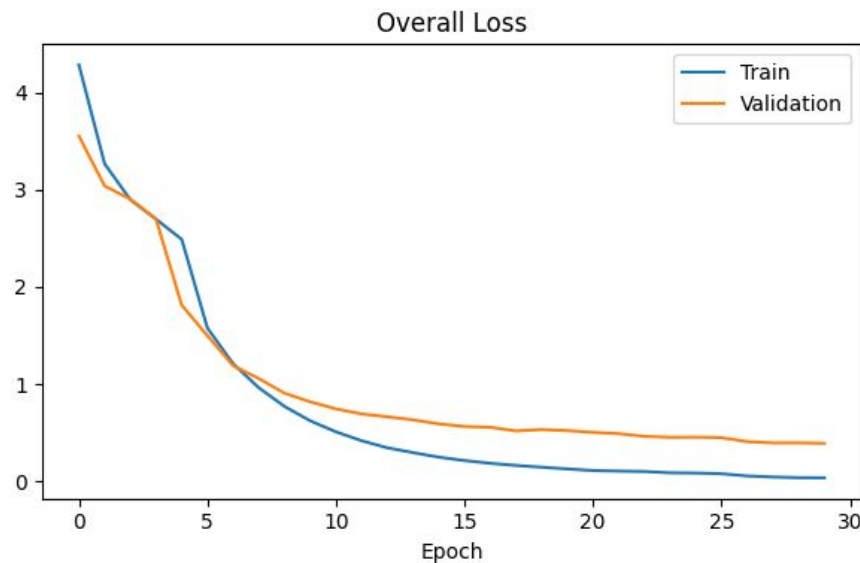


W-Specific Architecture

Multi-task vehicle verification model with 83 makes + 406 models outputs



Output	Classes	Random Top-1	Test Top-1	Loss
W model head	406	$1/406 = 0.25\%$	Acc = 95.15%	0.181
W make head	83	$1/83 = 1.20\%$	Acc = 95.99%	0.1486



Conclusion

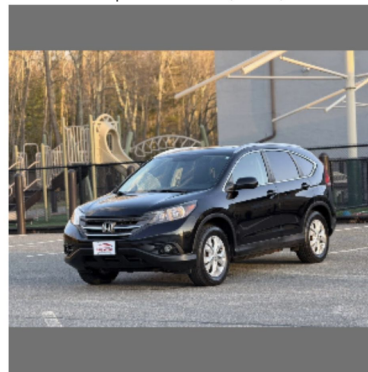
- v4ab: single-task 406-class model classifier using whole images
 - Test model Top-1: 91.99%
 - More directly deployable because no bbox is required
- W: multi-task classifier predicting both model and make
 - Model Top-1: 95.15%
 - Make accuracy: 95.99%
 - Higher performance, but depends on provided bbox crops
- Business takeaway: both models can support listing verification, but they fit different deployment settings
 - v4ab: user-uploaded marketplace photos
 - W: controlled settings or pipelines with detector/bbox support

Top-1: BMW 3 Series convertible (0.4676)



Top-5 predictions for 8136abc87b127bd9165d3830a65e94d4 (4).jpg:
1. BMW 3 Series convertible [81/86] prob=0.4676
2. BMW 5 Series estate [81/96] prob=0.1451
3. BMW 3 Series [81/68] prob=0.1117
4. BMW 5 Series [81/70] prob=0.0329
5. Ford New Focus hatchback [122/590] prob=0.0184

Top-1: Honda CR-V (0.9056)



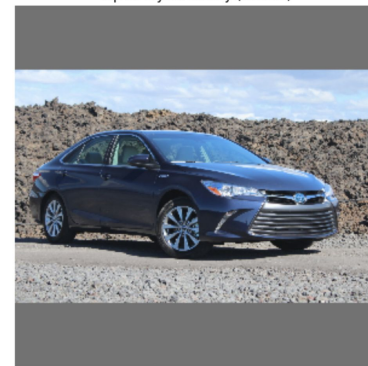
Top-5 predictions for 801476ac-f174-42db-af53-83ffaa4ad6b3.png:
1. Honda CR-V [100/252] prob=0.9056
2. Nissan Versa [97/883] prob=0.0084
3. Mitsubishi Zinger [28/1887] prob=0.0046
4. KIA SPORTAGE [140/1203] prob=0.0038
5. Mitsubishi i [28/1872] prob=0.0031

Top-1: Audi A6 (0.4676)



Top-5 predictions for audi_a6_sample.jpg:
1. Audi A6 [78/9] prob=0.4676
2. Audi A4L [78/2] prob=0.1451
3. Audi R8 [78/33] prob=0.1117
4. Audi A8L [78/27] prob=0.0329
5. BMW 5 Series [81/70] prob=0.0184

Top-1: Toyota Camry (0.1340)

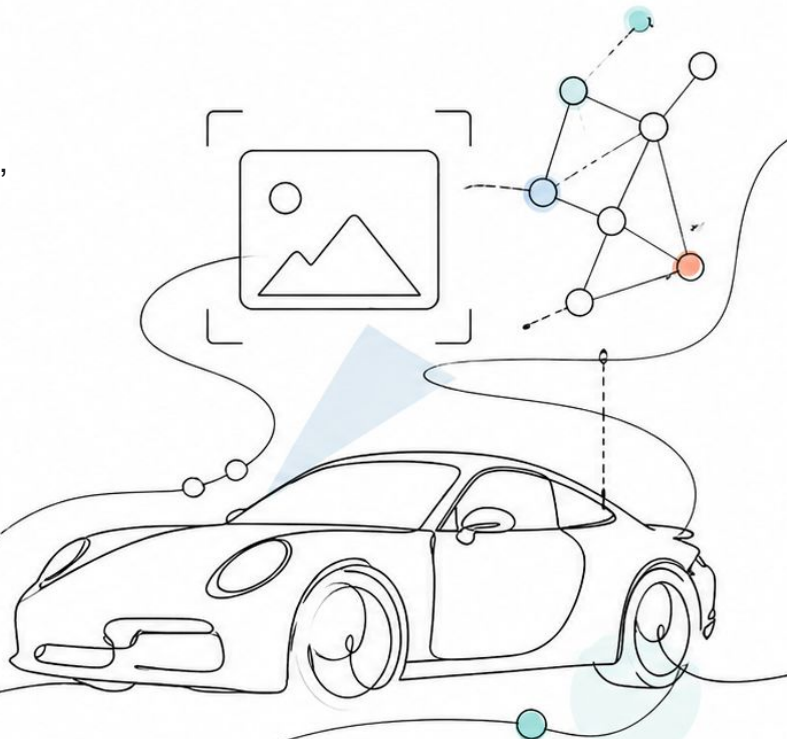


Top-5 predictions for 98fafc61ebec188a81f52c436cfb7aa8.jpg:
1. Toyota Camry [39/1386] prob=0.1340
2. Honda Accord [100/213] prob=0.0523
3. Toyota Reiz [39/1328] prob=0.0375
4. Toyota Corolla [39/1323] prob=0.0355
5. Lexus GS hybrid [159/1568] prob=0.0317

Next Steps

- **Deployable pipeline:** no-bbox training or detector + classifier
- **More realistic evaluation:** stricter splits, external test images, and newer vehicle classes
- **Business extension:** image-text mismatch flagging for fraud detection
- **Model update + pricing data:** retraining demo and unusual

price flagging



Thank you!

Appendix

```
Epoch 25: val_top1_acc improved from 0.77724 to 0.77724, saving model to outputs/best_model.keras - loss: 1.6598 - top1: 0.77724
Epoch 25: finished saving model to outputs/best_model.keras - loss: 1.6598 - top1: 0.77724, saving model to outputs/best_model.keras
Epoch 26: val_top1_acc improved from 0.77724 to 0.77724, saving model to outputs/best_model.keras - loss: 1.6598 - top1: 0.77724
Epoch 26: finished saving model to outputs/best_model.keras - loss: 1.6598 - top1: 0.77724, saving model to outputs/best_model.keras
Epoch 27: val_top1_acc improved from 0.77956 to 0.78992, saving model to outputs/best_model.keras - loss: 1.6386 - top1: 0.78992
Epoch 27: finished saving model to outputs/best_model.keras - loss: 1.6386 - top1: 0.78992, saving model to outputs/best_model.keras
Epoch 28: val_top1_acc improved from 0.78992 to 0.79322, saving model to outputs/best_model.keras - loss: 1.602 - top1: 0.79322
Epoch 28: finished saving model to outputs/best_model.keras - loss: 1.602 - top1: 0.79322, saving model to outputs/best_model.keras
```

Model Performance & Update Process

Key Model Changes

Version	Classes	Input Size	Data Split	BBox	Task
Baseline	336	224	Official + val carve	No	Single
v4a	336	299	Official + val carve	No	Single
v4ab	406	299	Custom 60/10/30	No	Single
Wilson	406 + 83 make	299	Stratified 70/15/15	Yes	Multi-task

[illegible]

Overview

Parameter	Initial train	v3	v4	v5	v4a	v4ab	W
Classes	336	336	336	336	336	406	406 model + 83 make
Input size	224	224	224	224	299	299	299
Batch size	32	32	32	32	16	16	32
Bbox crop	No	No	No	No	No	No	Yes (GT)
Task structure	Single (model)	Single	Single	Single	Single	Single	Multi-task (make+model)
Loss weighting	—	—	—	—	—	—	0.30 make + 0.70 model
Augmentation	None	RandAugment(num_ops=2, factor=0.5)	RandAugment(num_ops=1, factor=0.3)	RandAugment(num_ops=1, factor=0.3)	RandAugment(num_ops=1, factor=0.3)	RandAugment(num_ops=1, factor=0.3)	RandomFlip + Rotation(0.03) + Zoom(0.10) + Contrast(0.10)
RandomErasing	No	factor=0.5	factor=0.3	factor=0.3	factor=0.3	factor=0.3	No
MixUp	No	No	No	alpha=0.2	No	No	No
Bottleneck	None	256	512	512	512	512	512
Dropout	0.3	0.5	0.5	0.5	0.5	0.5	0.35
Unfreeze from layer	140	154	100	100	100	100	140
Label smoothing	0	0.1	0.1	0.1	0.1	0.1	0
Optimizer	Adam	AdamW	AdamW	AdamW	AdamW	AdamW	AdamW
Weight decay	0	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4
Normalization	/255	/255	/255	/255	/255	/255	resnet_v2.preprocess_input([-1,1])
LR schedule (finetune)	ReduceLROnPlateau	Cosine decay	Cosine decay	Cosine decay	Cosine decay	Cosine decay	ReduceLROnPlateau
Warmup LR	1e-3	1e-3	1e-3	1e-3	1e-3	1e-3	1e-3
Finetune LR	1e-4	1e-4	1e-4	1e-4	1e-4	1e-4	5e-5
Warmup / Finetune epochs	3 / 5	10 / 30	10 / 30	10 / 30	10 / 30	10 / 30	4 / 36
Early stop patience	4	7	7	7	7	7	6
Train/val/test split	Official + 15% val carve	same	same	same	same	Custom per-class 60/10/30 (seed 42)	Sklearn stratified 70/15/15 (seed 42)
MIN_IMAGES filter	None	None	None	None	None	≥120	≥120

Version	Val Top-1		Val Top-5		Train Top-1		Gap		Test Top-1		Test Top-5		Test Make Acc	Test Loss
Initial	46.29%	-	70.46%		99.67%	-	53.4%	-	0.5179	-	0.7576	-	-	2.1498
v3	56.81%	0.11	80.93%	0.10	57.10%	-0.43	0.67%	-0.53	0.5954	0.0775	0.8238	0.0662	Only Model	2.5796
v4	73.29%	0.27	89.73%	0.19	91.95%	-0.08	19.2%	-0.34	0.7290	0.2111	0.9049	0.1473	Implicit	2.0555
v5	69.48%	0.23	88.32%	0.18	78.40%	-0.21	9.0%	-0.44	0.7130	0.1951	0.9016	0.144	Learning	2.1126
v4a	82.62%	0.36	94.72%	0.24	94.88%	-0.05	12.5%	-0.41	0.8402	0.3223	0.9562	0.1986	Make	1.6888
v4ab	92.27%	0.46	98.33%	0.28	94.57%	-0.05	2.30%	-0.51	91.99%	0.402	98.04%	0.2228	-	1.4387
W	94.82%	0.49	98.79%	0.28	99.79%	0.00	4.97%	-0.48	95.15%	0.4336	99.21%	0.2345	95.99%	0.1720